

CHILDRESS, TX, USA

WMO: 723660

Lat: 34.427N

Lon: 100.283W

Elev: 1951

StdP: 13.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23007

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.7	21.0	1.6	6.4	30.3	5.8	7.9	32.7	31.3	50.0	27.6	49.7	9.2	10	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	102.3	70.4	99.7	71.0	97.3	70.9	76.5	91.8	75.3	90.5	74.1	89.5	13.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	128.8	82.7	71.1	123.4	81.2	70.0	118.8	80.3	41.3	91.9	40.0	90.5	38.9	89.3	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.8	24.6	21.3	DB	9.8	106.5	4.7	4.2	6.5	109.5	3.7	112.0	1.1	114.3	-2.3	117.4
			WB	8.3	78.1	4.2	2.2	5.2	79.6	2.7	80.9	0.4	82.1	-2.7	83.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	42.2	45.3	54.1	62.6	71.6	80.0	83.4	82.4	74.7	63.4	52.3	43.0
	DBStd	17.02	9.83	10.97	10.40	9.04	8.30	5.74	4.66	4.82	6.74	8.54	9.24	9.15
	HDD50	909	277	198	77	13	1	0	0	0	0	10	81	252
	HDD65	3016	708	554	360	147	33	0	0	0	10	132	388	684
	CDD50	5659	35	66	203	392	671	901	1036	1004	741	426	150	34
	CDD65	2290	0	2	22	75	238	452	571	539	301	83	7	0
	CDH74	28083	25	97	376	1049	2791	5452	7462	6565	3156	929	162	19
	CDH80	14812	2	23	115	428	1377	2975	4308	3704	1509	335	34	2
Wind	WSAvg	10.6	10.3	10.9	12.0	13.2	12.0	11.8	9.7	8.7	8.9	9.8	10.1	9.9
Precipitation	PrecAvg	20.2	0.7	0.9	1.3	1.9	3.3	3.3	1.5	2.3	2.5	1.6	1.1	0.7
	PrecMax	34.3	3.3	2.4	5.0	7.3	8.1	8.3	5.2	7.2	6.7	4.3	4.3	3.7
	PrecMin	12.6	0.0	0.0	0.0	0.1	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.7	0.8	0.7	1.3	1.6	2.2	2.3	1.3	1.6	2.0	1.4	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.6	83.7	89.1	95.3	102.2	106.9	105.5	105.3	100.2	92.7	84.5	77.0
		MCWB	53.3	55.7	59.5	59.6	65.0	68.9	71.0	70.4	70.3	65.0	58.6	54.0
	2%	DB	70.9	76.3	83.0	90.2	97.0	100.8	101.7	101.2	96.0	87.9	78.9	69.7
		MCWB	50.0	53.6	57.6	60.2	66.4	70.9	71.4	71.4	70.2	65.3	57.5	50.8
	5%	DB	64.4	70.0	78.1	85.0	92.5	97.0	99.3	98.6	92.4	83.6	73.5	63.6
		MCWB	47.4	51.2	56.9	60.3	66.9	70.7	71.8	71.9	69.8	64.2	55.6	48.4
	10%	DB	59.1	63.5	72.5	79.9	87.9	93.4	96.8	95.6	88.9	79.3	68.5	58.6
		MCWB	45.0	48.4	55.0	59.5	66.0	71.0	71.8	71.9	68.9	62.2	54.0	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	60.3	65.4	68.9	74.3	77.9	78.1	78.0	75.8	72.4	64.2	59.1
		MCDB	70.4	71.1	79.1	81.9	88.4	93.3	93.4	92.2	90.3	83.0	73.8	66.6
	2%	WB	52.0	56.0	62.9	66.8	72.4	75.9	76.3	76.1	73.7	69.7	61.1	53.4
		MCDB	66.6	70.7	74.5	78.3	86.2	91.1	91.9	91.3	87.6	80.6	71.1	65.8
	5%	WB	48.9	52.8	60.1	64.5	70.7	74.4	75.1	74.8	72.3	67.4	58.7	49.8
		MCDB	62.8	66.9	71.4	76.2	84.3	89.6	90.7	90.2	85.5	77.0	68.8	62.7
	10%	WB	45.9	49.6	57.0	62.2	69.1	73.1	73.8	73.5	70.9	65.2	55.9	46.5
		MCDB	58.0	62.9	69.6	74.1	82.8	88.2	89.8	89.0	83.5	74.4	66.8	57.5
Mean Daily Temperature Range	5% DB	MDBR	25.0	25.0	26.2	26.2	24.1	23.1	23.2	23.2	23.3	24.7	24.8	23.4
		MCDBR	34.7	35.9	34.0	33.7	30.7	28.1	26.9	27.3	28.0	30.4	33.1	32.0
		MCWBR	18.7	18.4	15.7	14.0	11.9	9.3	8.1	8.8	9.8	13.2	16.1	17.4
	5% WB	MCDBR	32.3	31.9	27.6	26.9	25.2	23.4	22.8	23.1	22.6	23.1	26.0	29.7
MCWBR		18.4	18.4	14.6	13.9	11.6	9.6	8.4	8.6	9.3	12.0	14.7	17.2	
Clear-Sky Solar Irradiance	taub	0.276	0.287	0.320	0.355	0.361	0.399	0.408	0.391	0.361	0.343	0.308	0.281	
	taud	2.558	2.535	2.450	2.368	2.424	2.349	2.378	2.377	2.440	2.464	2.499	2.547	
	Ebn at Noon	301	308	303	295	293	280	277	281	285	282	284	292	
	Edh at Noon	25	29	34	38	37	40	38	38	34	30	26	24	
All-Sky Solar Radiation	RadAvg	998	1214	1610	1951	2117	2290	2250	2038	1706	1326	1038	859	
	RadStd	92	145	100	143	205	167	111	109	101	166	119	72	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	N/A

Nomenclature: See separate page